

Explainable Prototype-Based Few-Shot Learning for Skin Lesion Classification

Ashish Yadav^{1†}, Bhavya Ahuja Grover^{1†}, Riya Bansal^{2†},
Nikhil Kumar Rajput^{1*†}

¹Ramanujan College, University of Delhi, Delhi, 110019, India.

²Department of Computer Science, University of Delhi, Delhi, 110007, India.

*Corresponding author(s). E-mail(s): n.rajput@ramanujan.du.ac.in;

Contributing authors: ashish.yadav.230104@gmail.com;

b.ahuja@ramanujan.du.ac.in; rbansal1@cs.du.ac.in;

[†]These authors contributed equally to this work.

Abstract

Deep learning has been highly successful in medical image analysis, but due to lack of availability of labeled data and lack of interpretability of such models, it has not yet been widely adopted in real-world clinical settings. Traditional convolutional neural networks (CNNs) are heavily dependent on large-scale annotated datasets which is scarce in medical and these models also work as opaque black-box models, so it makes them unsuitable for clinical applications and professionals where safety is top priority. In this paper, we present an explainable few-shot learning (XFSL) framework which integrates deep features learning, prototype-based classification, and similarity-based interpretability using the (Δ) metric, and this does not rely on computationally complex architectures or large-scale manual annotated data. The proposed framework is built upon the EfficientNet-B0 backbone, and then to learn domain-specific representations and also improving feature separability between visually similar classes (e.g., melanoma and benign nevi), it was fine-tuned and optimized with Focal Loss. Test-Time Augmentation (TTA) and Power Normalization were used to stabilize feature embeddings, which made the model more reliable from geometric perspective. Each query image was classified by measuring its similarity, by using cosine similarity. Beyond classical Grad-CAM visual explanations, we propose the Similarity-Gap (Δ) metric, for improved interpretability. This mathematically transparent metric measures the ambiguity in feature space between the top

competing predictions and provides a more interpretable alternative to overconfident probability based scores. We use this metric to implement an automated out-of-distribution detection mechanism that detects and rejects query images if they fall below the threshold. The CNN backbone of proposed framework shows a very reliable performance by achieving a peak validation accuracy of 81.22% and an ROC-AUC of 0.9511 on multi-class dermoscopic dataset. Under extreme data constraints, an N-shot ablation study identifies 20-shot learning as the optimal baseline, yielding a Mean Accuracy of 76.64% and a Macro F1-score of 0.7634, indicating a clinically conservative prediction behavior. Furthermore, the $\tau = 0.55$ threshold OOD strategy demonstrates strong generalization, successfully rejecting 94.12% of non-medical images and 82.35% of unseen clinical lesions. Overall, the results indicate that interpretable and uncertainty-aware classification can be achieved under severe data constraints, representing a step toward more reliable and interpretable AI systems in clinical decision support.

Keywords: Few-Shot Learning, Explainable AI, Medical Image Analysis, Deep Learning, Prototype-Based Learning, Grad-CAM, Computer Vision, Uncertainty Estimation

1 Introduction

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have significantly improved automated medical image analysis systems. These CNN models have shown strong performance in medical imaging tasks such as classification, detection, and segmentation tasks, in some cases they have produced expert-level detection [1, 2]. Even after such great achievements and success on many different levels, these models have not been widely adopted in real world medical settings because of many practical and complex methodological challenges.

The primary obstacle is the lack of large-scale labeled medical data. To get high quality annotations, it requires domain expertise which are costly in terms of person-hours, these are very time-consuming to generate, and also subject to strict privacy regulations. Due to this, many clinical datasets, especially for rare diseases or early-stage conditions, contain very few labeled samples per class [3]. In these cases, traditional deep learning models generally tend to overfit and have poor generalization performance.

The other big challenge is lack of interpretability. Most of the high performing deep learning models which have shown great results are black-box models with very little to almost no insight on how they reached to the final decision and their process involved in making that decision. In fields like healthcare, where safety is more important than anything else, this black-box nature of deep learning models reduce the trust of clinician and makes the regulatory approval even more strict and difficult [4, 5]. There is thus an increasing demand for models that are not only good in performance and accuracy but can also provide meaningful and reliable explanations.

To overcome this difficulty, Few-Shot Learning (FSL) has been developed as a promising approach for learning models that generalize from few examples helping

address the data scarcity problem. In particular, prototype-based methods, where models represent each class as a centroid in an embedding space and then classify query images based on similarity measures [6, 7]. While these approaches work effectively in settings where data is scarce, but these approaches typically do not have mechanisms to estimate prediction uncertainty or offer any valuable interpretable reasoning.

At the same time, many different Explainable Artificial Intelligence (XAI) techniques are being developed to make models more transparent and interpretable, Grad-CAM is one such example which have been developed to improve model’s transparency by highlighting regions of query image which significantly influence predictions [8]. However, these are mostly post-hoc methods and these are not inherently integrated into the decision making process of model, and therefore it limits their ability to give reliable estimations on uncertainty.

In this paper, we propose an explainable few-shot learning (XFSL) framework for medical image classification which helps in bridging the gap between data-efficient learning and interpretable decision making system. This proposed framework integrates deep feature extraction with prototype based classification, which makes effective learning in the limited data scenario possible. An EfficientNet-B0 backbone [9] which is then optimized with Focal Loss [10], and fine-tuned to extract domain-specific features while also proactively tackling feature-space collision between visually-similar classes. To make this feature extraction more stable, test-time augmentation (TTA) and power normalization were used before creating prototypes to the framework, without any data reduction in sample. Then we classify the query samples by cosine similarity in the optimized embedding space.

To make the framework more interpretable, reliable, and trustworthy, we introduce a similarity-based uncertainty estimation mechanism which evaluates the distance between top competing class predictions. This formulation which is referred as the Similarity-Gap (Δ), directly measures the ambiguity of the feature-space and then allows model to distinguish between the predictions whether they are confident or uncertain predictions. This mathematically transparent approach captures the geometric structure of the learned embedding space with more faithfulness as compared to other such approaches like overconfident softmax probabilities [11].

We propose a threshold based rejection mechanism which is based on this metric with empirically obtained threshold ($\tau = 0.55$) to safely filter out-of-distribution (OOD) samples [12]. This approach allows the system to choose whether to make a prediction or abstain from making any forced predictions on unfamiliar or ambiguous inputs which improves the safety of clinical deployment.

We tested the proposed framework on a multi-class dermoscopic dataset and it shows that proposed framework provides interpretable predictions while maintaining stable performance under limited-data scenarios.

After combining learning methods in settings with low data requirements with embedding-space uncertainty estimation, this framework gives a practical way towards the development of a more trustworthy AI system in healthcare. Importantly, this system also represents a clinically conservative diagnostic bias for dangerous skin lesions, with a greater tendency to recall cancerous than non-cancerous lesions for the sake of

precision, and this is consistent with the clinical safety margins. In contrast to other traditional methods of confidence estimation which need to be probability calibrated

1.1 Contributions

The key contributions of this work are as follows:

- **Explainable Few-Shot Learning Framework:** We have introduced a lightweight XFSL system which combines similarity-based interpretability using the (Δ) metric with prototype-based classification. This offers a transparent decision-making pipeline which is independent from big computationally demanding attention mechanisms or too costly human spatial annotations.
- **Improved Feature Separability:** Our proposed model have shown that during pre-training, the use of Focal Loss can successfully reduces the feature overlap between biologically similar lesion types (such as benign nevi and melanoma). This optimization technique gets a feature-space ROC-AUC of 0.9511, which shows that this optimization technique encourages distinct illness groupings.
- **Embedding Stabilization:** In order to compress extreme feature distributions and to improve 20-shot accuracy to 76.64% without requiring for aggressive outlier removal, we suggest combining Test-Time Augmentation (TTA) and Power Normalization.
- **Similarity-Gap (Δ) Metric:** Our suggestion is to quantify the geometric ambiguity between the top contending predictions using a feature-space uncertainty measure. This offers an alternative to overconfident softmax probabilities that is both theoretically straightforward and interpretable.
- **Clinically Conservative Diagnostic Bias:** We show that malignant recall is naturally prioritized in this framework over the benign precision, in the optimized 20-shot prototype space. Instead of forcing ambiguous lesions into false-negative benign classifications, this clinically conservative prediction tendency guarantees that they are treated with the proper clinical caution.
- **Uncertainty-Aware OOD Detection:** We develop an empirically validated rejection mechanism based on a cosine similarity threshold of $\tau = 0.55$. This mechanism demonstrates strong performance in filtering out-of-distribution samples, achieving up to 94.12% rejection of non-medical inputs and 82.35% rejection of unseen clinical lesions.

The remainder of this paper is organized as follows. Section 2 reviews related work in few-shot learning and explainable AI within the context of medical imaging. Section 3 describes the proposed explainable few-shot learning (XFSL) framework, including the Similarity-Gap (Δ) metric, out-of-distribution (OOD) detection mechanism, and Grad-CAM integration. Section 4 presents the dataset, experimental setup, and evaluation metrics, while Section 5 presents the N-shot ablation studies and overall evaluation results. Section 6 provides an in-depth analysis of the key findings, strengths, limitations, and potential future directions. Finally, Section 7 concludes the paper.

2 Related Work

2.1 Few-Shot Learning and Meta-Learning

Few-Shot Learning (FSL) aims to learn models capable of classifying novel categories using only a small number of examples. The primary objective is to learn discriminative feature representations that remain robust under extreme data scarcity [6, 7]. A significant portion of existing literature relies on meta-learning, where models are trained across multiple tasks to learn generalizable distance metrics or initialization parameters. Prototypical Networks [6] use a mean embedding (prototype) to represent each class and they perform classification by using Euclidean distance, and therefore enforcing intra-class compactness. Learnable similarity functions and quick gradient-based adaptation methods are further introduced by variants like Relation Networks [13] and Model-Agnostic Meta-Learning (MAML) [3].

In recent times, research has been turned toward Explainable Few-Shot Learning (XFSL) to overcome the inherent opacity of these models. For example, Wang et al. [14] proposed MTUNet, which works by highlighting informative picture of regions by using combinations of pairwise matching with a self-attention-based pattern extractor. Although this method achieved 5-way 1-shot accuracies of 58.13% on mini-ImageNet [15], also the study suggested that visual patterns can facilitate information transfer. However, these methods frequently rely on computationally demanding self-attention modules. Additionally, while these approaches offer visual interpretability, but they lack integrated mathematical procedures to address diagnostic ambiguity or quantify feature-space uncertainty without appreciably adding architectural complexity.

2.2 Few-Shot Learning in Medical Imaging

Some of the major domain-specific difficulties in medical imaging classification process include substantial intra-class variability, with severe class imbalance, and a lack of labeled data [16]. Additionally, to do data annotation, specific specialist knowledge and adherence to strict privacy standards are needed, which leads to the scarcity and datasets with extremely limited sample sizes. In recent times, by using direct expert assistance, several studies have tried to increase the spatial realism of FSL. Using a Grad-CAM-based Dice loss [8], Uddin et al. [17] developed a framework that integrates radiologist-provided regions of interest (ROIs) during training, increasing the accuracy from 77.09% to 83.61% on the BraTS (MRI) dataset. Similar to this work, Wang et al. [18] presented EGxFSL, which achieves good performance on MRI and chest X-ray datasets by combining spatial supervision with active learning. Other than these direct spatial supervision works, Nikpour and Armanfard [19] introduced LaHA which is a reinforcement learning based approach that uses Vision-Language Models (VLMs) as a reward signals for models to identify the "hard attention" patches.

Although these methods have greatly increased the spatial fidelity, but at the same time they also create some practical bottlenecks: they either rely heavily on the expensive, manually annotated expert spatial masks [17, 18], or they require computationally expensive reinforcement learning agents and external VLMs [19]. This shows

that the need of frameworks that can improve feature separability without the need of costly manual annotations or big architectural complexity.

2.3 Anomaly Detection and Reliability in Clinical AI

A very important requirement for the clinical adoption of AI systems is explainability [4, 20]. To provide safe deployment, models must be able to identify out-of-distribution (OOD) inputs from query in addition to being interpretable for safe use. Deviation Networks (DevNet), which learns end-to-end differentiable anomaly scores by enforcing statistically significant deviations for anomalous data, and these methods were used by Pang et al. [12] to address these issues of anomaly detection. Nevertheless, DevNet makes extensive use of complicated Multiple Instance Learning (MIL) formulas and artificial Gaussian priors.

Additionally, some other popular explainability techniques like Grad-CAM [8] and LIME [21] gives another approach in addition which offers qualitative spatial insights but they do not intrinsically quantify mathematical ambiguity between competing multi-class predictions, as noted by Lee et al. [22]. This gap emphasizes on the need of a lightweight, intrinsic safety mechanism which does not have to rely on artificial data distributions in order to construct a trustworthy OOD rejection mechanism directly within the embedding space and quantify feature-space ambiguity.

2.4 Identified Gaps and Context of This Work

A review of few literature have shown that existing XFSL approaches either rely on costly expert annotated ROI masks [17, 18] or computationally intensive reinforcement learning agents [19]. This work addresses three key gaps:

1. **Improved Feature Separability:** The common loss functions used in previous medical FSL research are inappropriate for the ability to distinguish between similar classes in the visual domain. Explicit attention to this problem of feature overlap is done in our proposed framework during pre-training of model by incorporating Focal Loss. Furthermore, we integrate Test-Time Augmentation (TTA) and Power Normalization to inherently stabilize these embeddings, eliminating the need for aggressive data-dropping heuristics.
2. **The Similarity-Gap (Δ):** Existing models rely heavily on overconfident softmax probabilities [4, 23]. We introduce a mathematically transparent feature-space Δ -metric to quantify diagnostic ambiguity between top competing class prototypes.
3. **Lightweight OOD Safety:** We introduce an empirically validated cosine-similarity thresholding mechanism ($\tau = 0.55$) that successfully rejects 94.12% of random non-medical images. Unlike complex binary anomaly detectors [12], this provides an inherent, computationally light safety mechanism for multi-class medical systems.

This proposed approach offers a more practical clinical decision-support framework by shifting the focus from purely visual heatmap explanations to quantifiable feature-space safety.

3 Methodology

The proposed framework integrates deep feature extraction, prototype-based Few-Shot Learning (FSL), and multi-layered interpretability into a unified medical image classification pipeline. The overall process comprises four key stages: (1) feature extraction using a convolutional neural network, (2) prototype construction from a restricted support set, (3) similarity-based classification of query samples, and (4) the generation of visual and reasoning-based explanations.

This work combines representational learning with similarity-based system, and which has resulted in more effective learnign and performance even in low data settings. The full architecture of this proposed system is shown in Fig. 1.

3.1 Feature Extraction using CNN

Feature extraction for images in this framework is done by using an EfficientNet-B0 architecture [9] which offers a trade-off of high representational ability and efficient computation. The network is fine-tuned on the pre-training database for the particular morphological characteristics of dermoscopic lesions in images provided during test, which then aligns with the more general objective of a lightweight architecture for possible use. The pre-training database is used to optimize the network for the particular morphological characteristics of dermoscopic lesions.

Ultimately, the network is reduced to a single classification layer and utilized as a feature extractor. To make sure the geometric stability of prototypes created for classes against morphological fluctuations, this system has used Test-Time Augmentation (TTA), which involves feeding forward the original image, horizontally inverted image, and vertically inverted image and averaging their subsequent outputs of images to model. The extreme feature-space values are then also compressed by taking the signed square root of the averaged vector – this is known as Power Normalization, or Tukey’s Transform. The resulting stable feature embedding is the function $f(x) \in \mathbb{R}^{1280}$, the embedding dimension. They are based on dense vectors which are used for the construction of prototypes and similarity-based inference.

3.2 Prototype-Based Few-Shot Learning

Prototype-based Few-Shot Learning (FSL), inspired by Prototypical Networks [6], represents each class as a central embedding. In the proposed framework, each class is represented by a single prototype defined from a small support set. Let $\mathcal{S}_c = \{x_i^c\}_{i=1}^K$ denote the support samples for class c , where K represents the shot size.

Since the CNN backbone is optimized using Focal Loss and the extracted features are further stabilized using TTA and Power Normalization, aggressive outlier filtering becomes unnecessary. Therefore, the prototype p_c is computed as the mean of all embeddings in the support set:

$$p_c = \frac{1}{K} \sum_{x_i \in \mathcal{S}_c} f(x_i) \quad (1)$$

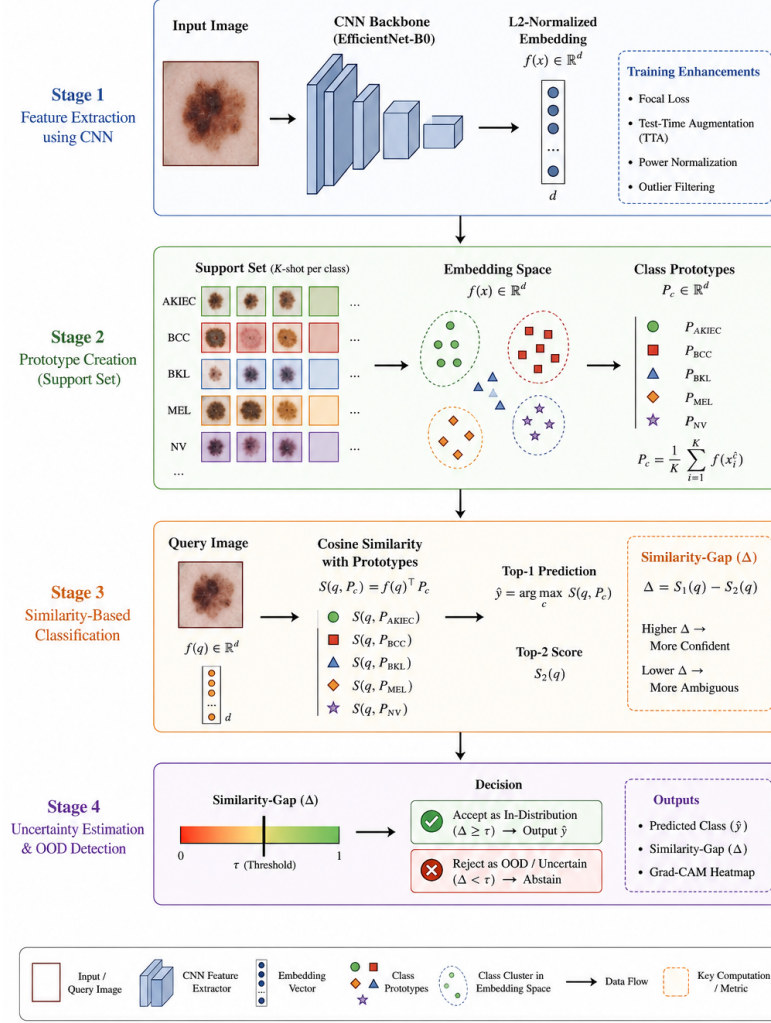


Fig. 1 Overview of the proposed XFSL framework for medical image classification. This pipeline consists of four main stages: (1) CNN-based feature extraction which was enhanced during training, (2) prototype creation from the support set for all classes, (3) similarity-based classification using cosine similarity and the Similarity-Gap (Δ) metric, and (4) uncertainty estimation with out-of-distribution (OOD) detection. This proposed framework integrates deep representation learning with interpretable and uncertainty-aware decision making.

The resulting prototype p_c and all query embeddings $f(x_q)$ are strictly L2-normalized to ensure stability and geometric consistency during cosine similarity computation.

3.3 Similarity-Based Classification

During inference, the feature embedding $f(x_q)$ of a query image x_q is extracted and compared against all class prototypes. Since both the query embeddings and the prototypes are L2-normalized, cosine similarity simplifies to the dot product:

$$S(x_q, p_c) = f(x_q) \cdot p_c \quad (2)$$

The predicted class $\hat{y}$ is assigned to the class with the highest similarity score:

$$\hat{y} = \arg \max_c S(x_q, p_c) \quad (3)$$

In addition to the primary prediction, the framework retains the second-highest similarity score. This enables quantitative analysis of prediction ambiguity and forms the basis for the uncertainty estimation mechanism described next.

3.4 Similarity-Gap (Δ) Ambiguity Quantification

Softmax probabilities [11] are frequently used by conventional neural networks to estimate confidence. But for most of the times, these methods are overconfident, giving false forecasts a high degree of certainty, which is especially troublesome in unclear medical situations.

We provide a similarity-based uncertainty measure that functions directly in the feature embedding space in order to overcome this constraint. Cosine similarity is calculated by using all class prototypes for a query sample. The greatest and second-highest similarity scores of samples are denoted by $S_{\text{top-1}}$ and $S_{\text{top-2}}$, respectively. The definition of the Similarity-Gap (Δ) is:

$$\Delta = S_{\text{top-1}} - S_{\text{top-2}} \quad (4)$$

The size of Δ is a direct measure of the reliability of prediction. A large Δ indicates high confidence, i.e. the predicted class is well separated from competing classes. On the other hand, a smaller Δ indicates an overlap in the embedding space, which corresponds to high diagnostic ambiguity.

In contrast to probability-based confidence scores, Δ is an explicit measure of feature-space separability, which is well aligned with learned representations. This yields a transparent and interpretable measure of uncertainty in prediction, which is important for reliable clinical decision making.

3.5 Unknown Detection (OOD Rejection)

To further improve system reliability, an uncertainty-aware rejection mechanism is incorporated into the decision layer. If the maximum similarity score for a query sample falls below a predefined threshold τ , the sample is classified as an out-of-distribution (OOD) [12] input. The prediction rule is defined as:

$$\hat{y} = \begin{cases} \text{UNKNOWN}, & \text{if } \max_{c \in \mathcal{C}} S(x_q, p_c) < \tau \\ \arg \max_{c \in \mathcal{C}} S(x_q, p_c), & \text{otherwise} \end{cases} \quad (5)$$

where $\mathcal{C}$ denotes the set of all classes.

The threshold τ was empirically found after doing ablation test on far and near OOD images with threshold settings of 0.5, 0.55, and 0.6 to balance in-distribution accuracy and anomaly rejection. After test we found 0.55 is more balanced and hence τ was set to 0.55. This mechanism acts as an automated safety layer which throws output as UNKNOWN for OOD images.

3.6 Explainability using Grad-CAM

Grad-CAM is used to create qualitative visual explanations in addition to quantitative uncertainty estimation provided by the Δ metric [8]. Grad-CAM calculates class-specific activation maps by taking gradients of class-scores in the activation map corresponding to the last convolutional layer.

Localization maps generated with these results are upsampled and overlaid onto the input image to show the regions with the strongest morphology that affect the prediction of the model (e.g., pigment networks, irregular borders). Further smoothing and blendings are used for visual clarity.

Both Grad-CAM and Δ contribute to the overall understandability of the decision-making process in this proposed dual-layer explainability framework, while the Grad-CAM feature provides spatial information about the model’s focus on the image, Δ mathematically quantifies the prediction ambiguity of model.

3.7 Inference Pipeline

Inference is the process where a query image is passed into the feature extractor which then returns the $L2$ -normalized image embedding for the query. These embeddings are then compared to all class prototypes present based on cosine similarity, and then it produces a ranked list of candidate diagnoses.

The final system output has three main elements: (1) the class label predicted using the maximum similarity, (2) the Similarity-Gap (Δ) value which reflects the reliability of prediction and (3) a heatmap of the clinically relevant spatial regions, which is obtained using Grad-CAM. Importantly, when the maximum similarity score is below the threshold $\tau = 0.55$ it is a safe interception and the input is classified as *UNKNOWN*.

4 Empirical Validation

In this section, the characteristics of the dataset, experimental setup and evaluation metrics for this study are described.

4.1 Dataset and Preprocessing

Experiments in this work are conducted on a subset of the HAM10000 dataset [16] which is available on the ISIC (International Skin Imaging Collaboration) archive and it serves as a widely used benchmark for skin lesion analysis. Five clinically relevant classes are evaluated: AKIEC (Actinic Keratoses and Intraepithelial Carcinoma), BCC (Basal Cell Carcinoma), BKL (Benign Keratosis), NV (Melanocytic Nevus), and MEL (Melanoma). The choice of these classes was made because they were chosen for being

of diagnostic importance and having a substantial amount of visual overlap, making this a stringent testbed for class performance and interpretability.

The dataset is split into two sets: one large pre-training set for CNN feature extraction, and a strictly balanced query set for the FSL evaluation. During CNN pre-training, the whole ISIC dataset is used without modifying the natural class imbalance from clinical practice which ranges from around 700 images with the rare diseases (e.g. AKIEC) to more than 12,000 images with common benign diseases (NV). This imbalance leads us to attempt to address the problem of the majority-class bias in representation learning by using Focal Loss [10] as a method.

The episodic FSL evaluation set is, on the other hand, made up of exactly 151 unseen images per class. Given the optimal value of $K = 20$, 20 images are randomly selected from the set as the support set with the remaining 131 images are used as the query set.

All images are resized to 224×224 pixels and normalized using ImageNet [15] mean and standard deviation values. To improve generalization and reduce overfitting, a robust data augmentation strategy is applied, including random 360-degree rotations, affine transformations (scaling and shearing), and color jittering. These augmentations encourage the model to learn orientation-invariant morphological features rather than relying on superficial artifacts.

4.2 Experimental Setup

The proposed model is evaluated on a five-class dermoscopic image dataset. The data is divided into a heavily imbalanced training set used exclusively to fine-tune the CNN backbone, and a perfectly balanced test set (151 images per class) used for few-shot evaluation. The EfficientNet-B0 [9] CNN backbone is fine-tuned for 8 epochs using the Adam optimizer. Due to the inherent difficulty of separating visually similar classes, such as Melanoma (MEL) and Benign Nevi (NV), Focal Loss [10] is utilized instead of standard cross-entropy. Focal Loss dynamically down-weights easy samples and focuses on hard-to-classify examples, thereby improving feature separation in overlapping regions of the embedding space.

In the Few-Shot Learning (FSL) evaluation, episodic support-query partitions are constructed. Specifically, K images per class are randomly sampled to construct class prototypes, and the remaining images are used as query samples. This process is repeated over 10 independent runs with random sampling to ensure statistical robustness.

4.3 Evaluation Metrics

The performance of the framework is evaluated using the following metrics:

- Training and Validation Accuracy
- Macro F1-Score and Matthews Correlation Coefficient (MCC)
- ROC-AUC (Embedding Space Separability)
- N-Shot Mean Accuracy and Standard Deviation
- Out-of-Distribution (OOD) Detection Rate

5 Results

5.1 CNN Training Performance

The CNN backbone performs well, as seen in Fig. 2 and summarized in Table 1.

Table 1 CNN Training Performance (Epoch 8)

Metric	Value
Final Training Accuracy	83.23%
Best Validation Accuracy	81.22%
Validation ROC-AUC	0.9511
Macro F1-Score	0.7358

This model has shown a stable and good performance on various metrics, like with a final training accuracy of 83.23% and then validation accuracy of 81.22% which also shows model is not overfitting and balanced. Also ROC-AUC is 0.9511 which shows that Focal Loss can reduce the feature overlap and make feature embedding space into separate clusters for further use.

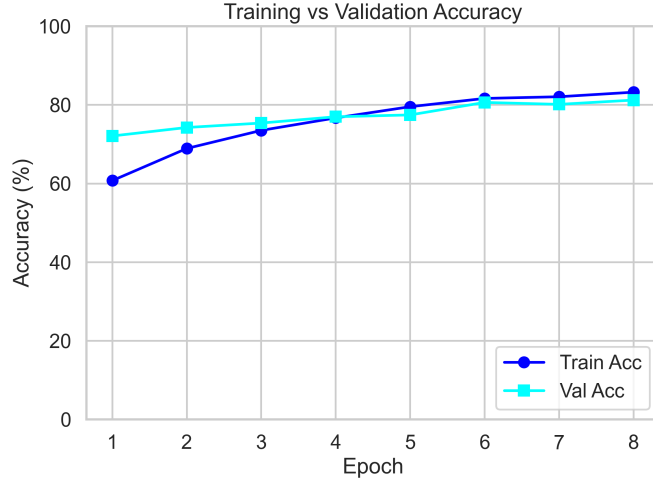


Fig. 2 Training and validation accuracy over 8 epochs show close alignment, indicating effective mitigation of overfitting.

5.2 XFSL Performance and N-Shot Ablation

To evaluate performance under constrained data conditions, an N-shot ablation study is conducted by varying the support set size K from 5 to 30. As shown in Fig. 3,

performance improves rapidly from $K = 5$ to $K = 15$, followed by a clear stabilization phase.

The best baseline of the framework is achieved at $K = 20$ with mean accuracy of 76.64%. It is also worth noting that the Macro F1-score (0.7634) is quite close to the accuracy, while the MCC reaches 0.7105. These results of experiment shows indication that all classes are predicted with similar accuracy and hence confirm the effectiveness of the combination of Focal Loss and embedding stabilization to reduce the bias of the majority class. Moreover, the performance robustness is shown by a small standard deviation of 1.23% over 10 runs.

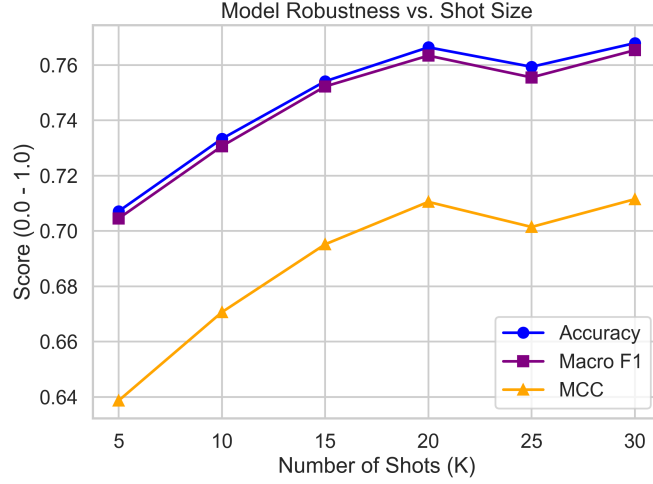


Fig. 3 Model robustness across varying support set sizes (K). Performance stabilizes at $K = 20$, establishing the optimal baseline.

Table 2 Few-Shot Learning Performance at Optimal $K = 20$

Metric	Value
Mean Accuracy	76.64%
Macro F1-Score	0.7634
Matthews Correlation Coefficient (MCC)	0.7105
Standard Deviation	1.23%
Number of Runs	10

When going beyond $K = 20$ we see diminishing returns (e.g. 75.94% for $K = 25$), which indicates that additional samples increase variability and noise in the prototype representations rather than enhancing them. And therefore the optimal support size for this dataset was observed at $K = 20$.

5.3 Per-Class Accuracy and Clinical Bias

Table 3 presents per-class performance at the optimal 20-shot setting. The results, supported by Fig. 4, highlight varying performance across morphological classes.

Table 3 Per-Class Performance at Optimal $K = 20$

Class	Precision	Recall	F1-Score
AKIEC	0.81	0.93	0.87
BCC	0.84	0.81	0.82
BKL	0.80	0.75	0.77
MEL	0.62	0.76	0.69
NV	0.87	0.64	0.74

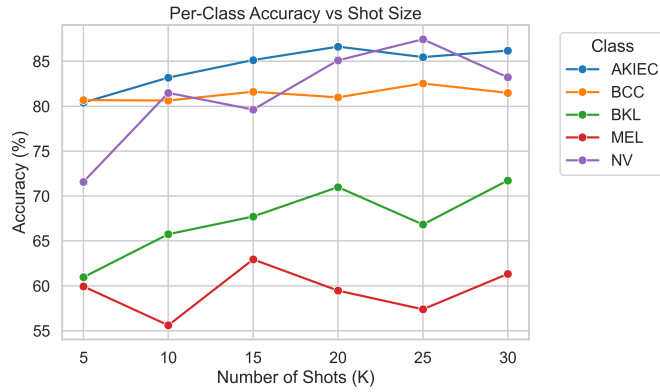


Fig. 4 Per-class accuracy across different N-shot support sizes. Melanoma (MEL) stabilizes near $K = 20$ due to feature-space optimization.

A key clinical observation is the model’s behavior for Melanoma (MEL) and Benign Nevi (NV). While NV shows high precision (87%), its lower recall (64%), combined with a Melanoma recall of 76%, indicates a conservative diagnostic bias. With a recall of 93%, the model achieves great sensitivity for different classes like AKIEC. The model prioritizes minimizing false negatives in malignant cases by cautiously classifying ambiguous benign lesions as potentially malignant.

5.4 Baseline Comparison

In order to compare the capabilities of the proposed XFSL framework with five existing classification baselines and understand its effectiveness after comparing with standard metrics, the performance of the same was compared under the same 20-shot setting. They encompass metric-learning methods (Vanilla Prototypical Networks,

Matching Networks and k-Nearest Neighbors) and conventional machine learning classifiers (Random Forest and Linear Support Vector Machine) while all using the same EfficientNet-B0 embeddings.

As demonstrated in Table 4 the proposed XFSL framework is competitive in its performance with other baseline methods, while simultaneously providing interpretability and uncertainty estimation. It attains a mean accuracy of 76.64% and a Macro F1-score of 0.7634. Existing methods based on a metric point of view like k-NN (71.63%) or Matching Networks (72.92%) show comparatively lower performance in the high dimensional embedding space, and Vanilla ProtoNet (75.89%) offers a good baseline. Test-Time Augmentation (TTA) is combined with Power Normalization to bring simple metric learning closer to more sophisticated classification methods.

Table 4 Performance Comparison Against Baselines at $K = 20$

Model	Mean Acc (%)	Macro F1	MCC
k-Nearest Neighbors ($k = 3$)	71.63	0.7103	0.6484
Matching Networks	72.92	0.7209	0.6694
Random Forest	75.07	0.7464	0.6901
Vanilla ProtoNet	75.89	0.7551	0.7008
Linear SVM	77.08	0.7667	0.7157
Proposed XFSL Framework	76.64	0.7634	0.7105

A Linear SVM works as a rigid classifier despite achieving a little greater raw accuracy (77.08%). In particular, SVM creates decision boundaries in the form of hyperplanes, which then requires a prediction for each input without a built-in rejection mechanism for query. This trend may lead to overconfident predictions on out-of-distribution (OOD) samples, which can be dangerous in clinical contexts.

However, this XFSL framework has given away very little 0.44% drop in raw accuracy but it has maintain well-bounded geometric prototypes of classes. A safety threshold ($\tau = 0.55$) and the computation of the Similarity-Gap (Δ) which is very crucial component of this framework, are both made possible by this design approach. This makes this paradigm more suitable for real-world medical applications, where precise uncertainty estimation is crucial. Fig. 5 provides visual support for this multi-metric performance profile, demonstrating the framework’s balanced stability across all evaluation criteria.

5.5 Feature Space Visualization (t-SNE)

To qualitatively validate the discriminative performance of this fine-tuned CNN backbone, t-Distributed Stochastic Neighbor Embedding (t-SNE) [24] is used to project the 1280-dimensional query set embeddings into a 2D space (see Fig. 6).

As per projection, the model maps dermatological morphological features to separate geometric clusters. For classes with clear separation, such as AKIEC and BCC, the margin between the classes is quite high. Melanoma (MEL) and Benign Nevi (NV)

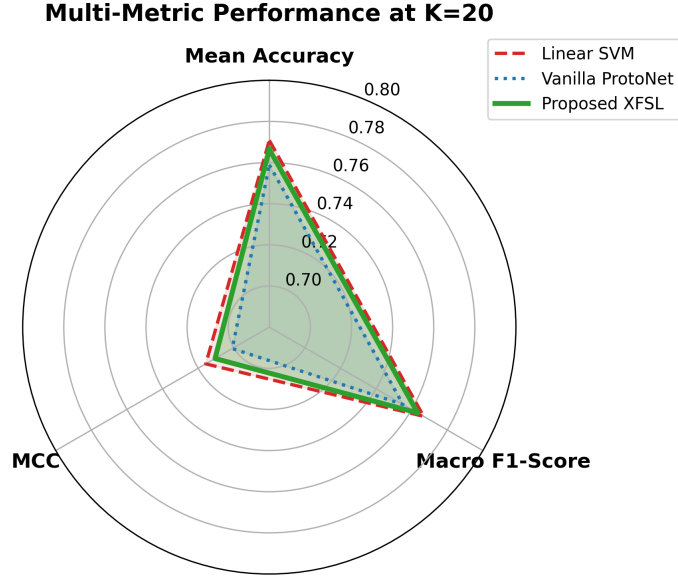


Fig. 5 Radar chart illustrating the multi-metric performance trade-off at $K = 20$. The proposed XFSL framework demonstrates a balanced geometric profile across Accuracy, Macro F1, and MCC, bridging the gap between simple metric learning (Vanilla ProtoNet) and rigid classifiers (Linear SVM).

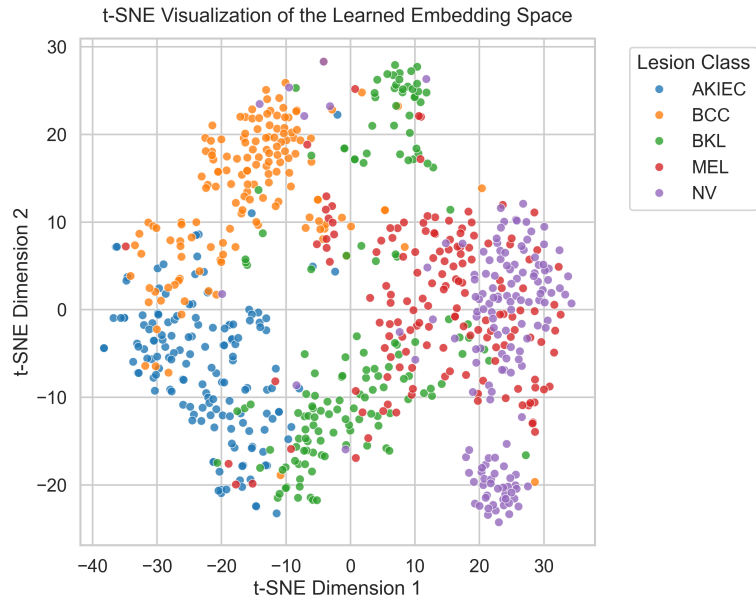


Fig. 6 t-SNE projection of the test set embeddings. The visualization shows strong geometric separation for distinct classes (e.g., AKIEC and BCC), along with expected biological overlap between Melanoma (MEL) and Benign Nevi (NV).

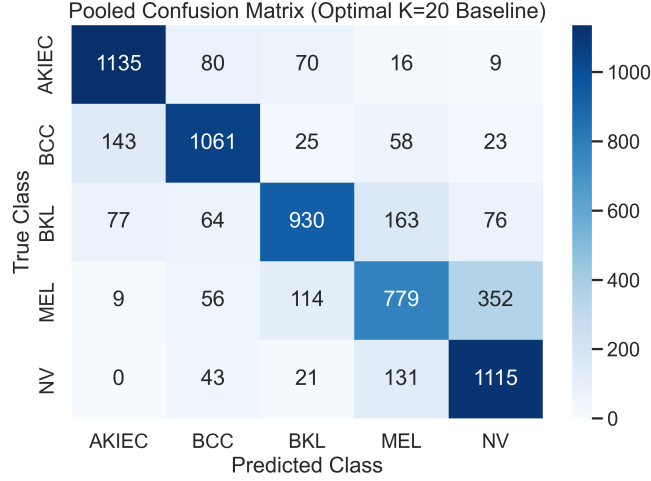


Fig. 7 Aggregate confusion matrix at $K = 20$. Misclassifications exhibit a conservative bias, prioritizing Melanoma (MEL) recall over benign precision.

are, however, near each other and exhibit a certain overlap in the embedding space. This observation is in accordance with known clinical features of dermoscopy, which indicates that the uncertainty of the model is not the result of a method limitation.

5.6 Confusion Matrix Analysis

To further analyze classification behavior, a pooled confusion matrix for the optimal baseline evaluation is generated. As shown in Fig. 7, most misclassifications occur between classes with high morphological similarity.

In particular, the confusion matrix reinforces the previously observed conservative bias. A significant portion of true NV (Benign Nevi) samples are classified as MEL (Melanoma). This is likely a consequence of clinical ambiguity and not a purely random error, which is implying that the model learns meaningful but overlapping features, and when presented with uncertainty, defaults to the clinically safer class.

5.7 Confidence and Similarity Gap Analysis

Similarity-based measures are analyzed to quantitatively assess prediction reliability.

As indicated in Fig. 8, predictions at low cosine similarity confidence generally tend to be mispredictions, but predictions at high confidence tend to be more widely dispersed. There is, however, still a clear overlap between these two distributions, implying that top-1 similarity confidence may not necessarily be sufficient to classify predictions into confident vs. questionable.

Based on this observation, it is important to use a supplementary uncertainty estimation measure such as the Similarity-Gap (Δ) metric. The distributions of

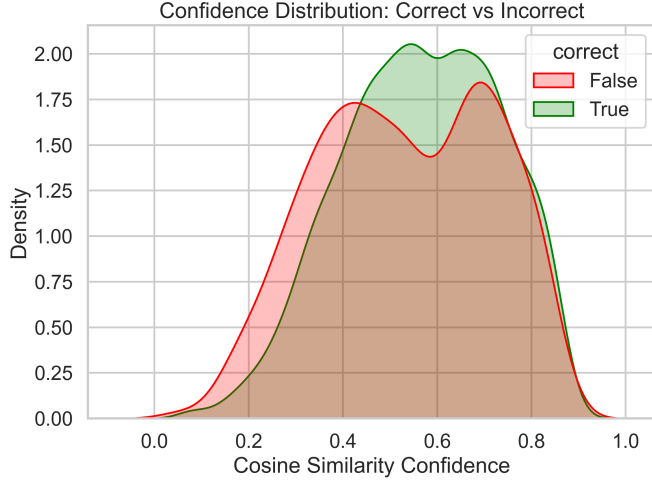


Fig. 8 Distribution of cosine similarity confidence scores for correct and incorrect predictions. Correct predictions generally exhibit higher similarity confidence values, while the incorrect predictions are mostly concentrated towards the lower confidence regions, although there remains a noticeable overlap between the two distributions.

Table 5 OOD Detection Rates across Thresholds (τ)

Threshold (τ)	Far-OOD (%)	Near-OOD (%)
0.50	86.27%	68.63%
0.55	94.12%	82.35%
0.60	98.04%	96.08%

similarity-gap (Δ) between the top first and second predictions are shown, and the similarities and dissimilarities are plotted for correctly classified and incorrectly classified cases (see Fig. 9).

The resultant distributions illustrate the effectiveness of the Δ metric. Many predictions are wrongly too close to 0.0, meaning that there is lot of uncertainty in the prediction space. However, with correct predictions there is a wider distribution associated with greater confidence and stronger geometric separation. This shows Δ to be a faithful and easy-to-interpret measure of prediction uncertainty that plays a similar role to overconfident softmax probabilities in [11].

5.8 Empirical Unknown (Out-of-Distribution) Detection

Next, we evaluate the framework for robustness and clinical safety using Out-of-Distribution (OOD) [12] samples with different threshold values. The evaluation dataset includes 51 random non-medical images (Far-OOD) and images of other unseen clinical lesion types which were not included in the training dataset such as DF, SCC, and VASC lesions (Near-OOD).

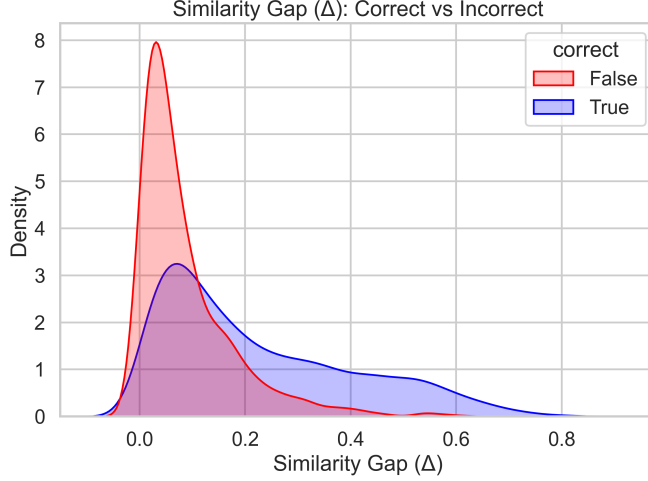


Fig. 9 Distribution of the Similarity-Gap (Δ). Incorrect predictions are strongly concentrated near 0.0, validating Δ as a reliable indicator of model uncertainty.

The model achieves a rejection rate of 94.12% of Far-OOD samples and 82.35% of dangerous Near-OOD medical samples at the empirically chosen threshold of $\tau = 0.55$. A sensitivity analysis shows that lower thresholds (e.g., $\tau = 0.50$) allow too many unseen samples to pass, while higher thresholds (e.g., $\tau = 0.60$) theoretically reject 98.04% of non-medical OOD samples but significantly increase the risk of incorrectly rejecting valid in-distribution images. Consequently, $\tau = 0.55$ is confirmed as the best trade-off between clinical safety and diagnostic usability. Sensitivity study shows that while higher levels (e.g., $\tau = 0.60$) theoretically reject 98.04% of non-medical OOD samples but significantly increase the chance of wrongfully rejecting genuine in-distribution photos, lower thresholds (e.g., $\tau = 0.50$) allow too many unseen samples to pass. The best trade-off between clinical safety and diagnostic usability is thus shown to be $\tau = 0.55$. The framework demonstrates strong safety boundaries, as detailed in Table 5 and visualized in Fig. 10.

5.9 Qualitative Results and Explainability

Qualitative analysis was performed using Grad-CAM visualizations to interpret the spatial focus of the model.

The generated heatmaps (Fig. 11) confirm that the model accurately focuses on clinically relevant regions of the lesion—such as irregular borders and pigment networks—substantiating the validity of the learned representations. When combined with the quantitative Δ metric and the *UNKNOWN* detection threshold, these visual explanations provide a comprehensive, multi-layered insight into both the model’s spatial focus and its underlying reasoning confidence.

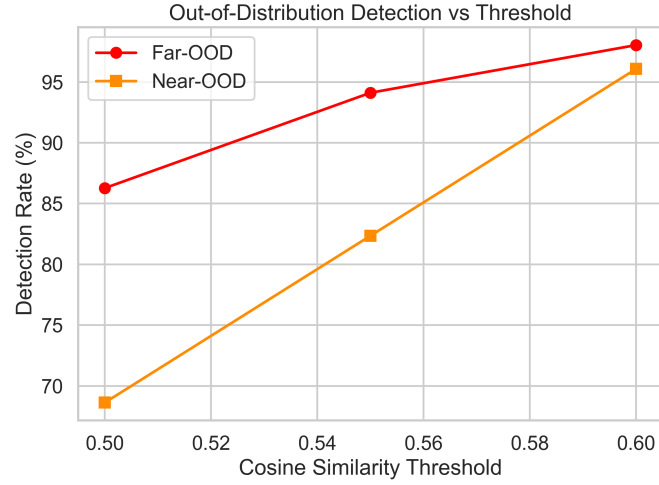


Fig. 10 Out-of-Distribution detection rates across varying cosine similarity thresholds. The $\tau = 0.55$ threshold balances optimal safety with clinical usability.

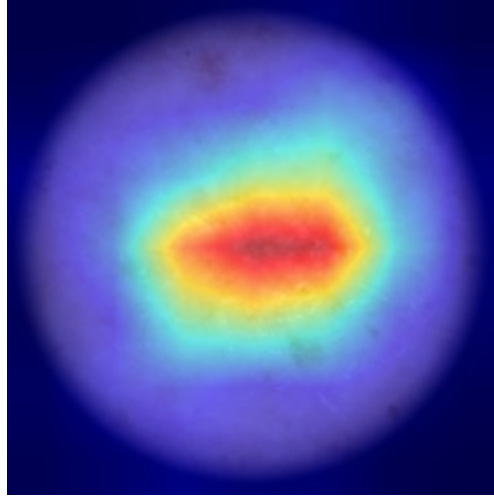


Fig. 11 Grad-CAM visualization highlighting the regions of the lesion that most strongly influenced the model's similarity prediction.

5.10 Summary of Findings

The experimental results highlight the following key observations:

- The CNN backbone provides well-separated feature representations, achieving an ROC-AUC of 0.9511.

- The prototype-based FSL approach maintains a stable 76.64% accuracy using only 20 samples, effectively improving feature separability between majority-class bias (Macro F1-score of 0.7634).
- Per-class analysis reveals a conservative, clinically conservative prediction tendency that prioritizes Melanoma recall over benign precision.
- The Similarity-Gap (Δ) metric provides a mathematically grounded and interpretable measure of prediction uncertainty.
- The empirically validated threshold $\tau = 0.55$ acts as an effective safety mechanism, rejecting up to 94.12% of unfamiliar inputs.

Overall, the framework successfully balances data efficiency, clinical safety, and interpretability.

6 Discussion

6.1 Analysis of Results

The experimental results have indicated that the proposed solution has been able to accommodate a compromise model of interpretability, data efficiency and accuracy. Despite the imbalance of the dataset, the CNN backbone can still obtain the discriminative representation of the features of various classes, yielding the highest accuracy of 81.22% on the validation set, and the area under ROC score curve of 0.9511.

Moreover, robustness of this proposed approach has been highlighted in the few-shot learning part. The average accuracy of the model stands at 76.64% which is at the optimal 20-shot configuration with a standard deviation of 1.23%. Furthermore, other indicators such as Macro F1-score (0.7634), MCC (0.7105) present the balance between the classes. This is a confirmation that the use of Focal Loss in combination with embedding stabilisation (employing TTA and Power Normalization) is a successful way to reduce the majority class bias.

Actionable behaviour is observed in challenging cases per class, as outlined in the per class analysis. There is feature space overlap between other classes, for example between Melanoma (MEL) and Benign Nevi (NV) with classes that do not do well, such as AKIEC. The model responds to this uncertainty conservatively with a very high (76%) recall rate, albeit at the cost of some benign, but low, precision. On average, this behavior indicates that there are fewer false negatives in malignant cases, thereby following clinical safety requirements, although a minor number of false positives may also occur, depending on the school size, teaching methods, and the specific disease.

6.2 Effectiveness of the Proposed Approach

Low data scenarios are a good fit for prototype-based classification using a deep feature extractor. The CNN backbone produces discriminative embeddings while Test-Time Augmentation (TTA) and Power Normalization ensure consistent and universal performance without requiring large datasets, balanced datasets, or severe outlier filtering.

Notably, the Similarity-Gap (Δ) metric gives a clear direction to think about uncertainty estimation. Δ is a real metric of the geometric distance in this embedding of

the class representations, not like softmax probabilities. We can see from the empirical distribution of Δ that the majority of the mispredicted data are concentrated around zero, thus allowing for the application of a criterion of ambiguity.

The uncertainty-aware rejection mechanism given in this model contributes more to the safety of system and this method. The model classifies low-similarity samples (i.e., $\max S(x_q, p_c) < \tau = 0.55$) as *UNKNOWN* which has a rejection rate of 94.12% for Far-OOD samples and 82.35% for unseen clinical lesions, and thus it avoids forced predictions on unfamiliar inputs.

The principal strengths of the proposed framework are summarized as follows:

- **Data Efficiency:** It demonstrates a competitive performance under low data constraints while using prototype-based Few-Shot Learning.
- **Class Balancing and Stability:** The integration of Focal Loss with Test-Time Augmentation (TTA) and Power Normalization has effectively mitigated the class imbalance and also morphological noise without removing data or filtering, achieving a robust Macro F1-score of 0.7634.
- **Uncertainty Estimation:** The Similarity-Gap (Δ) metric provides a mathematically grounded, feature-space-aligned measure of predictive reliability.
- **Robustness:** The threshold mechanism ($\tau = 0.55$) acts as a strict and automated safety boundary against out-of-distribution inputs.
- **Interpretability:** The synergistic combination of Grad-CAM with the Δ metric provides a cohesive visual and quantitative explanations for the query.

6.3 Comparison with State-of-the-Art Approaches

The suggested methodology shows that competitive performance can be obtained with a fraction of the annotated data as compared to traditional supervised CNN models. With just 20 support images per class, the 20-shot framework maintained 76.64% accuracy compared to the fully supervised baseline’s 81.22% accuracy. Fully supervised models continue to be inflexible, opaque black-boxes even if they perform somewhat better in data-rich circumstances.

More importantly, our approach has successfully overcome the major limitations of recent Explainable Few-Shot Learning (XFSL) frameworks. Recent medical imaging models achieve high spatial fidelity, but either they rely heavily on costly manually annotated expert ROIs [17, 18], or they require computationally intensive reinforcement learning agents [19]. Similarly, state-of-the-art Out-of-Distribution (OOD) detectors usually rely on complex synthetic prior generation [12].

In contrast, we gain separable disease clusters (AUC 0.9511), quantifiable uncertainty (Δ), and robust OOD rejection, just by means of embedding space mathematics and feature stabilization. It is an effective and scalable solution for medical applications where there are multiple competing problems to solve, including data scarcity and interpretation.

6.4 Limitations

The suggested framework has some drawbacks even after its advantages that call for more research and detailed evaluation. Firstly, the prototype-based formulation makes

the assumption that a single mean centroid in the embedding space may represent each class. Classes with substantial intra-class variability may be oversimplified by this assumption and such classes can have more variation and might need more prototypes, and therefore a multi-modal or distribution-based representations might more accurately portray the richness of the data.

Second, an out-of-distribution detection threshold $\tau = 0.55$ is set empirically for this dataset, it was obtained with testing other thresholds (e.g., 0.5 and 0.6) but it works on this dataset and may not adapt with this empirical setting on other datasets. The framework may need to be re-evaluated for other threshold or any automated detected threshold in other clinical databases or imaging modalities. Last, while Grad-CAM is valuable for giving spatial explanations, it is not necessarily clinically valid. The marked areas need to be checked against expert marked ground truth data before it can be used in the real world.

The evaluation is conducted on a subset of the HAM10000 dataset, which, although is widely used, but it represents a single imaging modality and clinical distribution. Therefore, the generalizability of the framework across other different clinical settings and imaging modalities (e.g., histopathology, radiology) remains to be tested.

6.5 Future Work

Future work will focus on improving both predictive performance and clinical interpretability. Potential directions include:

- Developing a multi-prototype or distribution-based representations to better capture intra-class variability which might need more support size for other prototypes.
- Incorporating a localized attention mechanisms to improve fine-grained feature discrimination in highly overlapping classes.
- Conducting quantitative evaluations with the help of expert-annotated bounding boxes to validate the clinical relevance of Grad-CAM heatmaps.
- Exploring adaptive and dynamically computed thresholding strategies for different datasets, cameras, and images to optimize out-of-distribution detection across different clinical environments.

7 Conclusion

In this paper, we introduce an explainable few-shot learning (XFSL) approach to tackle the two most important issues in medical image diagnosis when trying to adopt AI for this which are, restricted interpretability of deep learning models and data scarcity. The suggested method exhibits significant robustness under limited data conditions by combining geometric prototype-based classification with EfficientNet-B0 feature extraction. The framework successfully reduces morphological variances at the embedding level by utilizing Power Normalization and Test-Time Augmentation (TTA), which lessens the requirement for aggressive data-filtering techniques. In addition, we propose the Similarity-Gap (Δ) metric, a metric for prediction uncertainty in the embedding space, based on geometric considerations. This formulation

provides an interpretable basis for reasoning about diagnostic ambiguity, without the overconfidence problems of probability-based scoring methods.

Experimental results have shown that this proposed architecture has outperformed other few-shot methods on the 20-shot scenario with an average accuracy of 76.64%. This trade-off is relevant since the model trades one bit of raw accuracy to keep a bound of geometric prototypes, which is more suited to the application under consideration. Importantly the model makes a slight drop in raw accuracy in favor of bounded geometric prototypes that is more appropriate for the application being addressed. This design allows a similarity-based threshold for detecting and rejecting out-of-distribution (OOD) inputs be used with $\tau = 0.55$. In addition, the learned prototype space is also biased in a conservative direction for recalling malignant lesions, consistent with clinical safety considerations. The key takeaway from this work is the effectiveness of a representation learning framework based on data efficient representation learning, embedding stabilization, and embedding-space uncertainty estimation for medical decision support in practice, and its interpretability. In future, multi-prototype representations will be explored in order to better capture intra-class variability and adaptive uncertainty thresholds to be explored across larger multi-modal clinical datasets.

Declarations

- Funding: This work was not supported by any funding.
- Conflict of interest/Competing: The authors declare that they have no competing interests
- Data availability: Experiments are conducted on a subset of the HAM10000 dataset [16], available through the ISIC (International Skin Imaging Collaboration) archive.
- Author contribution: All authors contributed equally.

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